

Internalizing Environmental Risk: Insurance Design and Firm Behavior in Hazardous Industries

Kaleb K. Javier

June 22, 2026





- ▶ The firm pays for what it causes, regardless of fault — across oil & gas (OPA 1990), surface mining (SMCRA 1977), and hazardous substances (CERCLA 1980, RCRA 1976)
- ▶ Works when firms can be held fully accountable — breaks down when they cannot (Shavell 1986, 2007)
- ▶ Shavell (2004): two instruments to restore accountability:
 - ▶ Minimum Asset / bonding requirements
 - ▶ **Compulsory liability insurance** — where the premium does the deterrence work, *if* the insurer can price on risk-relevant attributes

This paper examines the design and efficacy of compulsory liability insurance as a financial responsibility instrument in environmental hazard settings.

Setting

- ▶ USTs — retail petroleum storage, $\sim$ 140,000 facilities, the most common source of groundwater contamination in the US (EPA 2024)
- ▶ Compulsory insurance the law since 1988 (RCRA Subtitle I, \$1M floor)
- ▶ **Clean shock:** Texas exits its flat-fee fund December 22, 1998 — 26,000 facilities enter risk-rated private insurance overnight
- ▶ 9.8M tank-years of EPA panel data spanning both regimes

Three questions

- ▶ **Q1 — Observability:** Is release risk predictable from observable tank characteristics?
- ▶ **Q2 — Behavior:** Does risk-based pricing change closure, replacement, and release outcomes?
- ▶ **Q3 — Welfare:** How much of the first-best welfare gain does risk-rated pricing capture?

[ADD FIGURE HERE]

Liability regimes, financial responsibility, and UST policy

Shavell (1986, 2004, 2007); Yin, Kunreuther & White (2011); Yin, Pfaff & Kunreuther (2011, 2007); Boomhower (2019); Ho, Hsu, Cha & Rivera (2018)

First firm-level panel analysis of insurance design — risk-rated vs. flat-fee — as a financial responsibility instrument; clean policy shock; structural welfare decomposition

Environmental economics and environmental risk

Fowlie, Reguant & Ryan (2016); Muehlenbachs (2015); Blundell, Gowrisankaran & Langer (2020); Kellogg & Reguant (2021)

Dynamic structural model of firm behavior under environmental regulation; welfare decomposition bounding gap to first-best

IO of selection markets: risk-based and cost-based pricing

Einav, Finkelstein & Mahoney (2021); Einav, Finkelstein & Cullen (2010); Bundorf, Levin & Mahoney (2012); Handel, Hendel & Whinston (2015); Mahoney & Weyl (2017)

[PLACEHOLDER]